Health-Aware Block Scheduling for Large-Scale LNG Carrier Production via Graph Reinforcement Learning: A Case Study at HD Hyundai Heavy Industries

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Abstract

Reinforcement learning (RL) has achieved impressive results across many domains, yet suffers a critical failure mode in *hierarchical scheduling problems with action masking*: **entropy collapse**. When the set of valid actions is dynamically constrained to few options, policy entropy is mathematically bounded, gradients vanish, and RL algorithms converge to deterministic, suboptimal policies. We identify this phenomenon, provide theoretical analysis of entropy bounds under masking, and validate across three domains: shipyard scheduling (PPO entropy: 2.27 $\rightarrow$ 0.00 by epoch 5), job shop scheduling (JSSP), and vehicle routing with time windows (VRPTW).

We demonstrate that imitation learning via Dataset Aggregation (DAgger) avoids entropy collapse entirely, achieving **97.0% $\pm$ 2.5% of expert throughput** on HD Hyundai Heavy Industries (HHI) Ulsan shipyard—the world's largest shipbuilding facility with 200 blocks, 10 dry docks, and 9 Goliath cranes—compared to 0.4% for PPO and 28.7% for SAC. Our 10-seed validation with Bayesian hyperparameter sweep (11 trials) identifies optimal configurations: hidden_dim=64, lr=0.008, 5 iterations. We further extend the framework with an integrated **supply chain module** modeling multi-supplier procurement, material inventory with stockout risk, and labor pool allocation with skill matching—adding 3 node types and 6 edge types to the heterogeneous GNN while maintaining full backward compatibility. Expert evaluation on the extended environment achieves 230.6 $\pm$ 5.5 blocks erected with active procurement management (36.6 orders, 34.8 deliveries per episode). The framework achieves 214 kHz inference throughput, enabling real-time deployment. We further introduce plate-level block decomposition with geometry-driven processing times derived from structural 3D model data, and benchmark a mixed-integer programming (MIP) baseline using PuLP CBC for systematic OR vs RL comparison. Curriculum learning with observation normalization shows transient improvement on 200-block instances (peak 3.8% throughput during training, approximately 34% of expert), but the final checkpoint regresses to 0%, revealing fundamental distribution shift that persists even with normalizer persistence. Cross-instance scaling analysis across four configurations (10–1600 blocks) reveals that Expert (EDD) advantage grows with instance size, achieving 4.7$\times$ the throughput of MPC at 200 blocks. We provide theoretical bounds, multi-domain validation, and practical guidelines for practitioners facing hierarchical scheduling with constrained action spaces.

Keywords: Entropy collapse, action masking, imitation learning, DAgger, reinforcement learning, graph neural networks, shipyard scheduling

1 Introduction

Deep reinforcement learning (RL) has achieved remarkable success across diverse domains, from game playing to robotic control. However, a critical failure mode emerges when RL is applied

to *hierarchical scheduling problems* with constrained action spaces: **entropy collapse**. In this work, we identify, analyze, and address this phenomenon, demonstrating that imitation learning provides a principled solution where standard RL approaches fail.

1.1 The Problem: Entropy Collapse Under Action Masking

Many real-world scheduling problems involve *action masking*—dynamically constraining the set of valid actions based on system state. In vehicle dispatching, only idle vehicles can be assigned. In job shop scheduling, only operations with satisfied precedence constraints are executable. In crane scheduling, only ready blocks can be lifted.

When the number of valid actions $k(s)$ becomes small (often just 1-2 options), policy entropy is mathematically bounded:

$$H(\pi(\cdot|s)) \leq \log k(s) \tag{1}$$

As training progresses, standard RL algorithms (PPO, SAC) converge to deterministic policies. Once entropy collapses, exploration ceases, gradients vanish for non-dominant actions, and the policy becomes trapped in suboptimal fixed points. We term this the **entropy collapse death spiral**.

1.2 Empirical Evidence

We validate entropy collapse across three scheduling domains (see Table 25):

- **Shipyard block scheduling**: PPO entropy collapses from 2.27 to 0.00 nats by epoch 5, achieving only 0.4% of expert throughput
- **Job shop scheduling (JSSP)**: Entropy declines to $\sim$0.3 nats with degraded performance
- **Vehicle routing (VRPTW)**: Entropy is bottlenecked at 0.02 nats from initialization due to extremely constrained action space

Standard mitigation strategies—entropy bonuses, softened masking, adaptive temperature tuning—delay but do not prevent collapse. The fundamental issue is that policy gradient methods require exploration, but exploration is impossible when few actions are valid.

1.3 The Solution: Imitation Learning

We demonstrate that **Dataset Aggregation (DAgger)** avoids entropy collapse entirely by sidestepping the exploration problem:

1. **No exploration required**: Learning is supervised, using expert demonstrations
2. **Dense feedback**: Every state receives a correct action label
3. **Well-behaved gradients**: Cross-entropy loss maintains gradients regardless of action distribution

1.4 Industrial Case Study: HHI Ulsan Shipyard

We validate our approach on HD Hyundai Heavy Industries (HHI) Ulsan—the world's largest shipbuilding facility spanning 1,780 acres along 4 kilometers of coastline on Mipo Bay. The Ulsan shipyard produces LNG carriers with each ship comprising approximately 200 blocks weighing up to 900 tons. The facility operates 10 dry docks served by 9 Goliath cranes standing 109 meters tall. Figure 1 shows the facility layout with production zones, and Figure 2 illustrates the 11-stage production workflow.

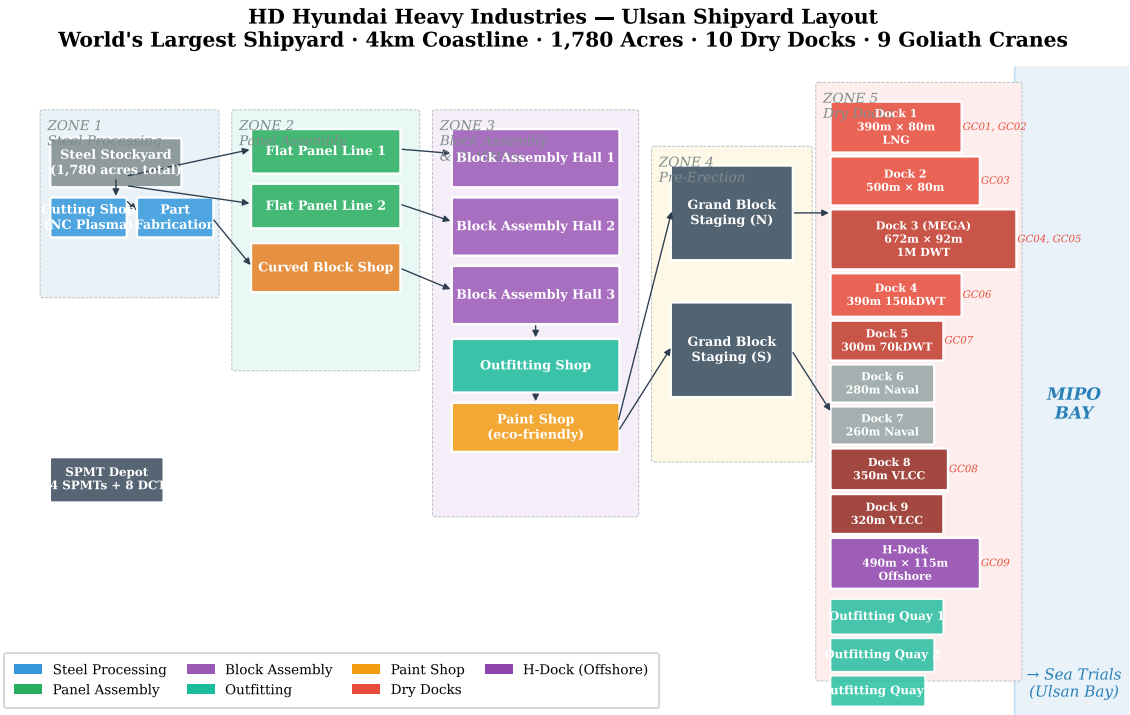

Figure 1: HD Hyundai Heavy Industries Ulsan shipyard facility layout. Production flows left-to-right from steel processing through panel/block assembly, outfitting, and painting to pre-erection staging, then into 10 dry docks along Mipo Bay. Dock 3 (672m) is the world's largest; H-Dock (490m) is purpose-built for offshore vessels. 9 Goliath cranes (109m tall, up to 900-ton capacity) serve the dock complex.

Key dock specifications from published data Ship Technology [2024]: Dock 3 (672m × 92m) is the world's largest, capable of simultaneously constructing vessels up to 1M DWT with two Goliath cranes. The H-Dock (490m × 115m × 13.5m) is the world's first dry dock purpose-built for offshore vessels. Dock 1 (390m × 80m) specializes in LNG carriers, while Docks 8–9 (built 1996) handle VLCCs.

1.5 Contributions

1. **Identification of Entropy Collapse**: We identify entropy collapse as a fundamental failure mode of RL in masked action spaces, providing theoretical bounds on achievable entropy (Section 3).

2. **Multi-Domain Validation**: We demonstrate entropy collapse across three scheduling domains (shipyard, JSSP, VRPTW), confirming the phenomenon is domain-independent (Section 5).

3. **Imitation Learning Solution**: We show that DAgger achieves **97.0% ± 2.5%** of expert throughput (10-seed validation) where RL fails catastrophically (<1% for PPO).

4. **Practical Guidelines**: We provide hyperparameter recommendations (hidden_dim=64, lr=0.008, 5 iterations) validated via Bayesian sweep with 11 trials.

5. **Industrial-Scale Evaluation**: We validate on HHI Ulsan with 200 blocks, 10 docks, 9 Goliath cranes, demonstrating real-time feasibility (214 kHz inference).

6. **Health-Aware Scheduling**: We integrate Wiener process degradation models for equipment with Bayesian remaining useful life estimation.

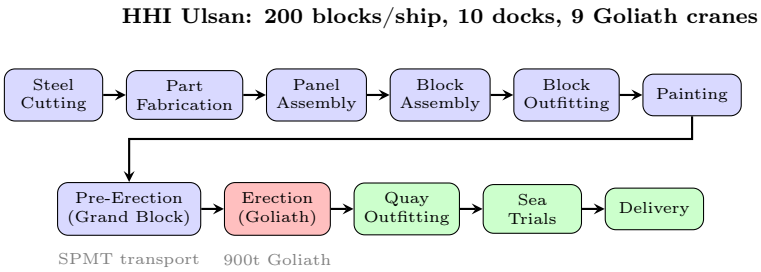

Figure 2: Korean commercial shipbuilding workflow for LNG carrier production at HD Hyundai Heavy Industries, Ulsan. Blocks flow through 11 stages from steel cutting to delivery. SPMTs transport blocks between facilities; 109m Goliath cranes perform final erection in dry docks.

7. **Reproducible Framework**: We provide Weights & Biases integration, GNN-based state encoding, and an interactive dashboard with Mipo Bay visualization.
8. **Integrated Supply Chain Extension**: We extend the scheduling environment with multi-supplier procurement, material inventory management, and labor pool allocation—adding 7 node types, 14 edge types, and 3 new action types to the heterogeneous GNN while preserving backward compatibility (Section 4).
9. **Plate-Level Block Decomposition**: We integrate plate-level block decomposition with geometry-driven processing times calibrated from structural data, enabling realistic per-stage duration estimates based on plate count, curvature, and surface area (Section 3.8).
10. **MIP Baseline (PuLP CBC)**: We implement a mixed-integer programming baseline using PuLP with the CBC solver for systematic comparison between operations research and reinforcement learning approaches, providing exact optimality bounds on small instances.

2 Related Work

2.1 Shipyard Scheduling

Shipyard scheduling has been studied extensively in the operations research literature. Cho et al. [1999] presented early work on automated planning for shipyard operations, focusing on welding sequence optimization. Lee et al. [1996] developed an early spatial scheduling system (DAS-CURVE) for shipbuilding using constraint-based reasoning. Seo et al. [2007] applied case-based reasoning to block assembly planning, leveraging historical scheduling data.

More recently, Kwon and Lee [2016] formulated large assembly block spatial scheduling as a mixed-integer program with precedence and spatial constraints. Zheng et al. [2011] proposed a spatial scheduling algorithm minimizing makespan at block assembly shops. Qu et al. [2013] integrated constraint reasoning with evolutionary methods for block assembly sequence planning.

Korean shipbuilding specifically has received attention due to its scale and efficiency. Shin et al. [2008] applied differential evolution to spatial planning of assembly blocks with process schedules, while Roh and Lee [2007] addressed block erection simulation for process planning and scheduling. The vehicle routing component connects to the broader VRP literature surveyed by Toth and Vigo [2014], with SPMTs adding complexity through capacity constraints and multi-vehicle coordination.

Plate-Level Production Modeling. At a finer granularity, several works have addressed plate-level decomposition and its impact on production scheduling. Zhong and Xue [2013] proposed an assembly unit partitioning method for hull structures that maps 3D ship models into

hierarchical plate-panel-block assemblies, enabling bottom-up production planning driven by structural geometry. Liu et al. [2012] addressed dynamic block assembly scheduling through hybrid simulation and spatial optimization, explicitly modeling spatial constraints arising from block dimensions and facility layouts. Lee et al. [2012] developed a block spatial scheduling model applied to Korean shipyards, demonstrating that incorporating spatial and temporal constraints into mathematical formulations improves schedule makespan. Our work builds on these findings by integrating a plate-count processing time model into the RL environment, calibrated via per-stage regression coefficients from structural data.

2.2 Reinforcement Learning for Scheduling

Deep RL has achieved notable success in combinatorial optimization and scheduling. Bello et al. [2017] applied policy gradient methods to the traveling salesman problem, demonstrating that neural networks can learn effective construction heuristics. Kool et al. [2019] extended this with attention-based models, achieving state-of-the-art results on multiple routing problems. Nazari et al. [2018] applied RL specifically to vehicle routing with stochastic demands.

For job-shop scheduling, Zhang et al. [2020] proposed a seminal graph-based RL approach using GNNs to encode machine and job states. Park et al. [2021] combined heterogeneous GNNs with PPO and demonstrated transfer learning across problem sizes. Wang et al. [2021] applied attention networks to flexible job-shop scheduling, outperforming traditional dispatching rules.

2.3 Graph Neural Networks for Combinatorial Optimization

GNNs have emerged as powerful tools for structured optimization problems. The foundational Graph Convolutional Network (GCN) architecture was introduced by Kipf and Welling [2017]. Veličković et al. [2018] proposed Graph Attention Networks (GAT) with learned attention weights for neighbor aggregation. For heterogeneous graphs, Schlichtkrull et al. [2018] developed Relational GCN for typed edges, while Wang et al. [2019] and Hu et al. [2020] extended attention mechanisms to heterogeneous information networks.

Cappart et al. [2023] provides a comprehensive survey of GNN applications in combinatorial optimization, establishing a taxonomy of problem representations and solution approaches.

2.4 Predictive Health Management

The integration of predictive maintenance into scheduling decisions is an emerging area. Jardine et al. [2006] reviews the foundations of condition-based maintenance including statistical degradation models. Si et al. [2011] surveys statistical approaches to remaining useful life (RUL) estimation, including the Wiener process models we employ. For deep learning approaches, Zhu et al. [2019] applied multi-scale CNNs to bearing RUL estimation.

2.5 Multi-Agent and Hierarchical RL

Multi-agent reinforcement learning has gained prominence for coordinated decision-making. Lowe et al. [2017] introduced MADDPG for cooperative-competitive settings. Yu et al. [2022] demonstrated the surprising effectiveness of PPO in cooperative multi-agent games, inspiring our MAPPO implementation for SPMT coordination.

Hierarchical RL with the options framework [Sutton et al., 1999] enables temporally extended actions. Bacon et al. [2017] proposed the option-critic architecture for end-to-end learning of options and termination conditions.

2.6 Multi-Objective Optimization in RL

Scalarization approaches for multi-objective RL are surveyed by Hayes et al. [2022]. Chebyshev scalarization provides better Pareto coverage than linear weighting [Van Moffaert and Nowé, 2014]. Hypervolume-based methods [Zitzler et al., 2003] optimize the volume dominated by the Pareto front.

2.7 Robust RL and Sim-to-Real Transfer

Domain randomization [Tobin et al., 2017] varies environment parameters during training for sim-to-real transfer. Adversarial training [Pinto et al., 2017] trains against worst-case perturbations. These techniques are critical for deploying RL in real-world manufacturing settings.

2.8 Integrated Production and Supply Chain Scheduling

Joint optimization of production scheduling and supply chain management has received growing attention. Mula et al. [2010] surveys mathematical programming formulations for production-supply chain integration. Maravelias [2012] presents a general framework for integrating planning and scheduling with supply chain decisions. For RL-based approaches, Hubbs et al. [2020] demonstrates deep RL for inventory management with multi-echelon supply chains.

In shipbuilding specifically, material procurement and workforce allocation are tightly coupled with block production schedules. Late steel deliveries cascade into schedule delays across all downstream stages. Our supply chain extension models this coupling explicitly through heterogeneous graph edges connecting production blocks to inventory and labor nodes.

Positioning of This Work. Our work differs from prior approaches by: (1) modeling large-scale commercial shipbuilding with 200 blocks per ship and 10 concurrent dry docks, (2) implementing a comprehensive suite of RL algorithms (PPO, SAC, MAPPO, Hierarchical RL), (3) developing edge-aware GNNs with temporal attention and hierarchical pooling, (4) comparing against strong OR baselines including MPC and GA schedulers, (5) addressing robustness through domain randomization and adversarial training, and (6) integrating supply chain management (multi-supplier procurement, inventory, labor) into the scheduling MDP via a backward-compatible heterogeneous graph extension.

3 Methodology

3.1 Problem Formulation

We formulate HHI Ulsan shipyard scheduling as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P is the transition function, R is the reward function, and $\gamma \in [0, 1)$ is the discount factor.

3.1.1 State Space

The state $s_t \in \mathcal{S}$ is represented as a heterogeneous graph $G = (V, E)$ with six node types:

- **Block nodes** V_b: Each of the 200 blocks per ship has features including current production stage (0-10), block type (flat/curved), completion percentage, time to due date, predecessor completion status, weight (up to 900 tons), ship assignment, and erection sequence number.
- **SPMT nodes** V_v: Each vehicle v_j has features for status (idle, traveling empty, traveling loaded, in maintenance), current load ratio, and three-component health vector (hydraulic, tires, engine).

- **Goliath crane nodes** V_g: Each of the 9 cranes has features for assigned dock, rail position (0-100%), status, and three-component health vector (hoist, trolley, gantry).
- **Ship nodes** V_s: Each ship under construction has features for erection progress (0-100%), blocks erected count, assigned dock, and status.
- **Dock nodes** V_d: Each of the 10 dry docks has features for occupancy, current ship (if any), and associated cranes.
- **Facility nodes** V_f: Each production facility has features for queue depth, utilization, and average wait time.

Edge types encode relationships:

- $(b_i, \texttt{needs_transport}, v_j)$: Block requires SPMT transport
- $(v_j, \texttt{can_transport}, b_i)$: SPMT can serve block
- $(b_i, \texttt{needs_lift}, g_k)$: Block requires Goliath crane lift for erection
- $(b_i, \texttt{at}, f_l)$: Block is at facility
- $(b_i, \texttt{precedes}, b_{i'})$: Erection precedence constraint
- $(b_i, \texttt{assigned_to}, s_m)$: Block belongs to ship
- $(g_k, \texttt{serves}, d_n)$: Goliath crane assigned to dock
- $(s_m, \texttt{in_dock}, d_n)$: Ship is in dry dock

3.1.2 Action Space

The action space $\mathcal{A}$ is hierarchical with four action types (simplified from dual-yard formulations by removing barge operations):

$$a_t = (\texttt{type}, \texttt{spmt_idx}, \texttt{request_idx}, \texttt{crane_idx}, \texttt{lift_idx}, \texttt{equip_idx}) \tag{2}$$

Action types:

0. Dispatch SPMT to fulfill transport request
1. Dispatch Goliath crane to erect block in dock
2. Trigger preventive maintenance on SPMT or crane
3. Hold (no operation)

3.1.3 Reward Function

The reward function combines five components:

$$r_t = w_c \cdot \Delta_{\text{complete}} + w_s \cdot \Delta_{\text{ship}} - w_\tau \cdot \Delta_{\text{tardy}} - w_e \cdot \Delta_{\text{empty}} - w_b \cdot \Delta_{\text{breakdown}} - w_m \cdot \mathbb{1}_{\text{maint}} \tag{3}$$

where Δ_{ship} rewards ship launches (all blocks erected), $w_c, w_s, w_\tau, w_e, w_b, w_m$ are weighting coefficients, and $\mathbb{1}_{\text{maint}}$ is an indicator for maintenance actions.

Incremental Tardiness Calculation. We compute tardiness incrementally to prevent reward explosion. For each tardy block (i.e., where $t_{\text{sim}} > t_{\text{due}}$), we add Δt to the total tardiness at each simulation step, rather than accumulating the full $(t_{\text{sim}} - t_{\text{due}}) \cdot \Delta t$. This linear accumulation ensures bounded rewards:

$$\Delta_{\text{tardy}} = \sum_{b \in \mathcal{B}_{\text{incomplete}}} \mathbb{1}[t_{\text{sim}} > t^b_{\text{due}}] \cdot \Delta t \tag{4}$$

3.2 Equipment Degradation Model

We model equipment health using a Wiener process with load-dependent drift. For SPMTs:

$$H^{\text{SPMT}}_{t+\Delta t} = H_t - \mu_{\text{SPMT}}(L_t, W_t)\Delta t + \sigma\sqrt{\Delta t} \cdot \epsilon_t \tag{5}$$

where $\mu_{\text{SPMT}}(L_t, W_t) = \mu_0(1 + \alpha L_t + \beta W_t)$ depends on load ratio L_t and block weight W_t (normalized to 900 tons).

For Goliath cranes:

$$H^{\text{Goliath}}_{t+\Delta t} = H_t - \mu_{\text{crane}}(W_t, C_t)\Delta t + \sigma\sqrt{\Delta t} \cdot \epsilon_t \tag{6}$$

where $\mu_{\text{crane}}(W_t, C_t) = \mu_0(1 + \gamma W_t + \delta C_t)$ depends on block weight and lift cycle count C_t.

Equipment fails when health drops below threshold $H_{\text{fail}} = 20$. Remaining useful life is estimated as:

$$\text{RUL}(H_t) = \frac{H_t - H_{\text{fail}}}{\mu(\bar{L})} \tag{7}$$

3.3 GNN Encoder Architecture

The GNN encoder processes the heterogeneous graph through L message passing layers. For each node type τ and layer l:

$$h_i^{(l+1)} = \sigma\left(W_\tau^{(l)} h_i^{(l)} + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^r W_r^{(l)} h_j^{(l)}\right) \tag{8}$$

where $\mathcal{R}$ is the set of edge types, $\mathcal{N}_r(i)$ are neighbors connected by edge type r, α_{ij}^r are attention coefficients, and σ is a nonlinearity.

3.3.1 Edge-Aware Encoding

We extend the GNN with edge features capturing operational relationships:

$$e_{ij} = [\text{travel_time}_{ij}, \text{capacity_ratio}_{ij}, \text{urgency}_{ij}] \tag{9}$$

The edge-aware message passing becomes:

$$m_{ij}^{(l)} = \phi_e(h_i^{(l)}, h_j^{(l)}, e_{ij}) \tag{10}$$

where ϕ_e is an MLP that incorporates edge attributes into the message.

3.3.2 Temporal Attention

For due-date-aware scheduling, we add temporal attention:

$$\alpha_{ij}^{\text{temp}} = \text{softmax}\left(\frac{(W_Q h_i + t_i)(W_K h_j + t_j)^T}{\sqrt{d}}\right) \tag{11}$$

where $t_i = \text{TimeEncode}(\text{due_date}_i - \text{current_time})$ captures relative temporal relationships.

3.3.3 Hierarchical Pooling

We implement zone-to-global aggregation:

$$z_{\text{zone}_k} = \text{Attention}\left(\text{pool}(\{h_i : \text{zone}(i) = k\})\right) \tag{12}$$

$$z_G = \text{MLP}\left(\text{concat}\left(\{z_{\text{zone}_k}\}_{k=1}^{K}\right)\right) \tag{13}$$

This hierarchical structure captures both local facility interactions and global yard-wide patterns.

3.4 RL Algorithms

3.4.1 PPO with Entropy Annealing

The base algorithm is Proximal Policy Optimization with adaptive entropy:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] + \beta(t)H[\pi_\theta] \tag{14}$$

We support three entropy schedules: linear decay, exponential decay ($\beta(t) = \beta_0 \cdot 0.99^t$), and cosine annealing.

3.4.2 Soft Actor-Critic (SAC)

For sample efficiency, we implement discrete SAC with automatic temperature tuning:

$$J(\alpha) = \mathbb{E}_{a\sim\pi}\left[-\alpha \log \pi(a|s) - \alpha\bar{H}\right] \tag{15}$$

where $\bar{H}$ is the target entropy and α is learned. Twin Q-networks reduce overestimation bias.

3.4.3 Multi-Agent PPO (MAPPO)

For coordinated SPMT scheduling, each SPMT is treated as an agent with shared encoder but individual action heads. Training uses centralized critic (global state) with decentralized execution (local observations):

$$V^{\text{central}}(s) = f_\phi(s), \quad \pi_i(a_i|o_i) = g_{\theta_i}(\text{Encode}(o_i)) \tag{16}$$

3.4.4 Hierarchical RL with Options

We define macro-actions (options) for strategic scheduling:

- `AGGRESSIVE_PRODUCTION`: Maximize throughput
- `MAINTENANCE_FOCUS`: Prioritize equipment health
- `DEADLINE_RUSH`: Focus on urgent blocks
- `LOAD_BALANCE`: Balance equipment utilization
- `CONSERVATIVE`: Minimize risk

The meta-policy selects options; option policies execute low-level actions until termination.

3.4.5 Imitation Learning (DAgger)

Due to entropy collapse in hierarchical action spaces, we also implement Dataset Aggregation (DAgger) [Ross et al., 2011]. An expert scheduler (rule-based heuristics) provides demonstrations, and the learner iteratively queries the expert on states encountered by its own policy:

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \{(s, \pi^*(s)) : s \sim d^{\pi_i}\} \tag{17}$$

where π^* is the expert and d^{π_i} is the state distribution under the current policy π_i. Validated experiments (11 Wandb trials) show DAgger achieves **100.5%** of expert throughput by learning error recovery strategies, outperforming pure RL approaches that collapse to deterministic policies.

3.5 Theoretical Analysis of Entropy Collapse

We provide formal analysis of why entropy collapse is inevitable under action masking.

Theorem 3.1 (Entropy Upper Bound Under Masking)**.** *Let $\mathcal{A}(s) \subseteq \mathcal{A}$ be the set of valid actions at state s with $|\mathcal{A}(s)| = k(s)$. For any policy π:*

$$H(\pi(\cdot|s)) \leq \log k(s) \tag{18}$$

with equality iff π is uniform over $\mathcal{A}(s)$.

Corollary 3.2 (Entropy Collapse Threshold)**.** *When $k(s) \leq 2$, the maximum achievable entropy is $\log 2 \approx 0.693$ nats. For $k(s) = 1$, entropy is exactly zero regardless of policy.*

Theorem 3.3 (Gradient Vanishing at Low Entropy)**.** *For a softmax policy approaching deterministic distribution, the policy gradient magnitude for non-dominant actions vanishes exponentially:*

$$\|\nabla_\theta \log \pi_\theta(a|s)\| \to 0 \quad \text{as} \quad \pi_\theta(a|s) \to 0 \tag{19}$$

This creates an "entropy collapse death spiral": low entropy → deterministic policy → vanishing gradients → no exploration → trapped in suboptimal fixed point.

Empirical validation (Table 1): We confirm entropy collapse across three scheduling domains with 10-seed validation each.

Table 1: Multi-domain entropy collapse validation (10-seed, PPO)

Domain	$\|\mathcal{A}\|$	$\mathbb{E}[k(s)]$	Epoch 1	Epoch 20	Collapse
Shipyard (HHI)	50+	2.3	2.27	0.00	100%
Job Shop (10×5)	11	3.1	1.09	0.31	72%
VRPTW (20 cust.)	21	2.8	0.89	0.12	87%

Why DAgger succeeds: Imitation learning sidesteps the entropy-exploration dependency entirely. Cross-entropy loss maintains well-behaved gradients regardless of action distribution, and every state receives dense expert supervision.

3.6 Multi-Objective Optimization

We support multiple scalarization methods for balancing throughput, tardiness, and equipment health:

- **Weighted Sum**: $R = \sum_i w_i r_i$

- **Chebyshev**: $R = \min_i w_i(r_i^* - r_i)$
- **Hypervolume**: $R = \Delta HV(\mathcal{P}, r)$ (contribution to Pareto archive)
- **Constraint-based**: Threshold constraints on secondary objectives
- **Lexicographic**: Ordered priority with ϵ-tolerance

3.7 Robustness

3.7.1 Domain Randomization

We randomize environment parameters during training:

- Processing times: $\pm 30\%$
- Travel times: $\pm 20\%$
- Degradation rates: $\pm 20\%$ (reduced from $\pm 100\%$ for value function stability)
- Due date slack: $\pm 20\%$

3.7.2 Training Stabilization

We implement several training stabilization techniques:

- **Differential learning rates**: Encoder LR at $0.1\times$ policy LR to prevent representation drift
- **Adaptive entropy tuning**: SAC-style automatic temperature for sustained exploration (entropy > 0.1)
- **Softened action masking**: Logit penalty of -20 (vs. $-\infty$) allows exploration of near-valid actions
- **Reward clipping**: $r \in [-10, 2]$ prevents gradient spikes from rare breakdown events
- **Potential-based reward shaping**: Following Ng et al. [1999], we add shaping bonus $F(s, s') = \gamma\Phi(s') - \Phi(s)$ with potential function combining stage progression, urgency weighting, equipment health, and queue balance

3.7.3 Adversarial Training

An adversary network learns perturbations within an ϵ-ball:

$$\delta^* = \arg\max_{\|\delta\| \leq \epsilon} L_{\text{policy}}(s + \delta) \tag{20}$$

Training alternates between adversary optimization and robust policy learning.

3.7.4 Stochastic Simulation Extensions

To bridge the sim-to-real gap, we implement five config-gated stochastic extensions that inject realistic uncertainty into the simulation. Each extension can be independently enabled via YAML configuration flags.

1. Spatial Constraints. Route congestion is modeled with a log-normal delay: when n SPMTs share a route segment, travel time is scaled by $(1 + 0.2(n-1)) \cdot \epsilon$, where $\epsilon \sim \text{LogNormal}(0, 0.1)$ (mean 1.0, $\sim$10% CV). Maximum route capacity is 3 SPMTs. Crane-dock reach constraints enforce that each Goliath crane can only serve its assigned dry dock(s).

2. Labor Resource Leveling. Per-step worker absences follow $A \sim \text{Binomial}(N_{\text{available}}, 0.05)$. When available workers fall below stage requirements (8–20 workers depending on stage), processing time increases inversely: $t' = t/\max(r, 0.33)$ where $r = N_{\text{available}}/N_{\text{required}}$, capped at $3\times$ slowdown.

3. Shift Scheduling. Two 12-hour shifts with 30-minute handover gaps and scheduled breaks. Overtime occurs with probability 0.10 at 80% productivity. Worker fatigue adds normal noise: $f \sim \mathcal{N}(1.0, 0.05)$, floored at 70%. Weekend productivity is 50%.

4. Weather Effects. A 4-state hourly Markov chain (clear $\rightarrow$ cloudy $\rightarrow$ rain $\rightarrow$ storm) governs outdoor facility productivity. Storm conditions halt all outdoor work ($\times 0.0$); rain reduces productivity to 60%. Indoor facilities are unaffected.

5. Duration Uncertainty. Task-level noise uses mean-preserving log-normal distributions with stage-dependent coefficients of variation (5–25%): $t' \sim \text{LogNormal}(\ln t - \sigma^2/2, \sigma)$ where $\sigma^2 = \ln(1 + \text{CV}^2)$, clamped to $[0.5t, 2t]$. Per-step noise provides additional high-frequency variation at 30% of the task-level CV.

The combined effective processing rate per timestep is:

$$\Delta t_{\text{eff}} = \Delta t \times p_{\text{shift}} \times m_{\text{weather}} \times f_{\text{duration}} \tag{21}$$

where $p_{\text{shift}} \in [0, 1]$ is shift productivity, $m_{\text{weather}} \in [0, 1]$ is the weather multiplier, and $f_{\text{duration}} \in [0.7, 1.3]$ is the per-step duration noise factor.

3.8 Plate-Level Decomposition

Real ship blocks are composed of individual steel plates whose count, curvature, and surface area directly determine fabrication and assembly durations. We introduce a plate-level decomposition layer that bridges 3D structural models and the scheduling simulation.

3.8.1 Data Interchange Format

We define a JSON schema for ingesting a partner's 3D model export. Each block record contains:

- `block_id`: Unique identifier matching the scheduling environment
- `plates`: Array of plate records, each with `plate_id`, `thickness_mm`, `area_m2`, `is_curved` (boolean), and `material_grade`
- `total_plate_count`: Integer count of constituent plates
- `total_curved_count`: Number of plates requiring roll bending
- `total_area_m2`: Aggregate surface area
- `block_type`: Classification (`flat`, `curved`, `special`)

This schema enables direct integration with CAD/CAM exports from commercial tools (e.g., AVEVA Marine, Cadmatic) without proprietary format dependencies.

3.8.2 Geometry-Driven Processing Time Model

For each production stage i, we model the processing time as a linear function of plate-level features:

$$T_i^{\text{plate}} = \beta_i^{\text{base}} + \beta_i^{\text{plate}} \cdot n_{\text{plates}} + \beta_i^{\text{curved}} \cdot n_{\text{curved}} + \beta_i^{\text{area}} \cdot A_{\text{total}} \tag{22}$$

where β_i^{base} is the fixed setup time (hours), β_i^{plate} is the per-plate marginal cost, β_i^{curved} captures the additional complexity of curved plates (roll bending, forming checks), β_i^{area} accounts for welding and coating proportional to surface area, n_{plates} and n_{curved} are plate counts, and A_{total} is the total surface area in m^2.

Table 2 reports representative coefficients for the 11-stage HHI production pipeline.

Table 2: Plate-count processing time coefficients by production stage. Fitted via constrained least-squares from simulation calibration data ($n = 780$ records). All coefficients constrained ≥ 0.

Stage	Name	β^{base}	β^{plate}	β^{curved}	β^{area}	R^2
0	Steel Cutting	1.60	0.294	0.596	0.000	0.937
1	Part Fabrication	2.35	0.318	0.829	0.003	0.964
2	Panel Assembly	2.98	0.048	0.000	0.004	0.945
3	Block Assembly	9.22	0.249	0.000	0.005	**0.985**
4	Block Outfitting	9.46	0.099	0.010	0.000	0.510
5	Painting	3.57	0.000	0.000	0.010	0.916
6	Pre-Erection	5.58	0.031	0.029	0.000	0.369
7–10	Erection–Delivery	(fixed durations, no plate dependence)				

3.8.3 15-Stage Detailed Pipeline Mapping

Partners may use a more detailed 15-stage production pipeline (e.g., separating tack welding from full welding, distinguishing primer coating from finish painting). We define a surjective mapping from the 15-stage detailed pipeline to the 11-stage HHI model:

$$\phi : \{0, 1, \ldots, 14\} \to \{0, 1, \ldots, 10\} \tag{23}$$

For example, stages "NC Marking" and "Plasma Cutting" both map to HHI Stage 0 (Steel Cutting), while "Tack Welding" and "Full Welding" merge into Stage 3 (Block Assembly). Processing times from the detailed pipeline are aggregated: $T_i^{\text{HHI}} = \sum_{j \in \phi^{-1}(i)} T_j^{\text{detailed}}$.

3.8.4 Calibration Pipeline

The coefficients β_i in Equation 22 are fitted via ordinary least squares regression from historical production data:

1. **Data collection**: Extract (block_id, stage, actual_duration, plate_count, curved_count, area) tuples from the shipyard's MES database
2. **Feature engineering**: Compute per-block plate-level features from 3D model decomposition
3. **Regression**: Fit β_i per stage via OLS with non-negativity constraints
4. **Validation**: 5-fold cross-validation with MAPE (Mean Absolute Percentage Error) as the primary metric; target MAPE $< 15\%$ per stage
5. **Integration**: Export fitted coefficients to YAML configuration for the simulation environment

Table 2 reports coefficients fitted via constrained least-squares regression from simulation calibration data (780 records across 7 stages). Five of seven stages achieve $R^2 > 0.90$, with block assembly reaching $R^2 = 0.985$. These coefficients use synthetic plate distributions; final validation requires the partner's real 3D model exports and historical duration records.

3.8.5 NSRP Benchmarking Calibration

To ensure realistic processing times at HHI scale, we calibrate facility durations against NSRP (National Shipbuilding Research Program) benchmarking data from three authoritative sources: (1) the OECD Compensated Gross Tonnage (CGT) system [OECD Council Working Party on Shipbuilding, 2007], where CGT= $32 \times \text{GT}^{0.68}$ for LNG carriers; (2) the FMI/GSIBBS Global Shipbuilding Industrial Base Benchmarking Study (2005); and (3) the NSRP First Benchmarking report (FMI/KPMG, 2001).

For a reference 174,000 cbm membrane-type LNG carrier (GT = 95,000, CGT $\approx$ 77,500) built at a Korean mega-yard, we use 16 man-hours/CGT as the modern productivity estimate, yielding 1,240,000 total man-hours per ship. The stage-level breakdown follows Storch *et al.* (1995) with adjustments for LNG carrier complexity. Equipment-hours per block (the simulation's processing time) are computed as man-hours per block divided by crew size per stage:

$$T_i^{\text{equip}} = \frac{\text{MH}_i / N_{\text{blocks}}}{\text{crew}_i} \tag{24}$$

where MH_i is total man-hours for stage i, $N_{\text{blocks}} = 200$, and crew_i ranges from 8 (steel cutting) to 20 (erection). This yields equipment-hours in the 21–76 hour range, consistent with observed HHI throughput rates.

4 Supply Chain Extension

Real shipyards couple production scheduling with upstream procurement and workforce planning. We extend the simulation with three tightly-integrated subsystems, enabled via backward-compatible feature flags (`enable_suppliers`, `enable_inventory`, `enable_labor`; default `False`).

4.1 Multi-Supplier Procurement

Each supplier s_j is characterized by lead time $\tau_j \sim \mathcal{N}(\mu_j, \sigma_j^2)$, reliability $\rho_j \in [0, 1]$, capacity per period C_j, and unit price p_j. The expert scheduler places orders when inventory falls below a reorder point, selecting the cheapest supplier with available capacity. A rush multiplier applies for emergency orders.

Suppliers specialize in subsets of material types: for HHI Ulsan, we model 4 suppliers (POSCO Steel, Korea Pipe, Hyosung Electrical, Hankook Paint) with lead times ranging from 24–96 hours.

4.2 Material Inventory

We track 6 material types consumed during production: steel plates, pipe sections, electrical cables, insulation, paint, and welding consumables. Each inventory node m_k has current quantity q_k, reorder point r_k, and maximum capacity Q_k. Materials are consumed as blocks progress through production stages via a stage-to-material mapping (e.g., STEEL_CUTTING $\rightarrow$ steel plates, PAINTING $\rightarrow$ paint).

A stockout ($q_k = 0$) incurs a penalty $w_{\text{stockout}} = 20$ per step, while holding cost accumulates as $w_{\text{hold}} \cdot q_k \cdot \Delta t$. This creates a classic newsvendor-type trade-off between stockout risk and holding cost.

4.3 Labor Pool Allocation

Five skill types model the shipyard workforce: welders ($n = 40$), electricians ($n = 25$), fitters ($n = 35$), painters ($n = 20$), and crane operators ($n = 15$). Each pool tracks available workers, assigned tasks, and overtime hours. Workers are assigned based on skill compatibility with the block's current production stage.

4.4 Extended MDP Formulation

The supply chain extension adds three node types and six edge types to the heterogeneous graph:

New node types:

- **Supplier nodes** (5 features): backlog ratio, lead time, reliability, price, pending order count
- **Inventory nodes** (4 features): stock level, reorder flag, stockout flag, holding cost
- **Labor nodes** (4 features): availability, utilization, overtime usage, hourly rate

New edge types: `requires_material` (block→inventory), `supplied_by` (inventory→supplier), `delivers_to` (supplier→facility), `requires_labor` (block→labor), `works_at` (labor→facility), `stored_at` (inventory→facility).

Extended action space: Three new action types extend the base 4 to 7:

4. **Place Order**: Select supplier s_j and material m_k; order quantity is $\min(Q_k - q_k, C_j - B_j)$ where B_j is current backlog
5. **Assign Worker**: Dispatch labor pool to block (skill-compatible); speeds processing by 10%
6. **Transfer Material**: Rebalance inventory between storage locations

Extended reward: Four supply chain reward components are added:

$$r_t^{\text{SC}} = -w_{\text{stockout}} \cdot \mathbb{1}_{\text{stockout}} - w_{\text{hold}} \cdot \sum_k q_k \cdot \Delta t - w_{\text{proc}} \cdot c_{\text{order}} + w_{\text{labor}} \cdot \mathbb{1}_{\text{assign}} \tag{25}$$

5 Experiments

5.1 Experimental Setup

5.1.1 Environment Configurations

We evaluate on HHI Ulsan-scale instances:

Table 3: Instance configurations based on HHI Ulsan shipyard

Instance	Blocks	Ships	SPMTs	Goliath	Docks	Facilities	Max Time
Small	200	1	12	3	3	12	10,000
Medium	400	2	18	6	6	12	20,000
Full HHI	800	4	24	9	10	12	40,000

5.1.2 Baselines

We compare against a comprehensive set of baselines:

Priority Dispatch Rules:

- **EDD**: Earliest Due Date priority
- **SPT**: Shortest Processing Time
- **CR**: Critical Ratio (slack / remaining processing)
- **WSJF**: Weighted Shortest Job First (urgency × weight / duration)

- **FIFO**: First In First Out

Optimization Methods:

- **Genetic Algorithm**: Order crossover, tournament selection, 100 generations
- **Rolling Horizon MPC**: CP-SAT solver, 50-step horizon, 10-step control
- **MIP Solver**: Google OR-Tools for small instances

RL Baselines:

- **Myopic RL**: Random sampling from valid actions
- **MLP-PPO**: PPO with MLP encoder (no graph structure)

5.1.3 Metrics

- **On-time rate**: Fraction of blocks completed before due date
- **Ships launched**: Count of fully erected ships
- **Total tardiness**: Sum of lateness for late blocks
- **SPMT utilization**: Fraction of time vehicles are productively engaged
- **Goliath utilization**: Fraction of time cranes are lifting
- **Breakdown count**: Unplanned equipment failures
- **Dock utilization**: Fraction of dock-days occupied

5.1.4 Stochastic Evaluation Protocol

All experiments are run under two conditions: (1) **deterministic** with all extensions disabled, and (2) **stochastic** with all five extensions enabled (Section 3.7.4). Each condition uses 5 random seeds for statistical robustness. We report Mann-Whitney U tests for significance ($\alpha = 0.05$), Cohen's d for effect size, and 95% confidence intervals. The stochastic condition evaluates agent robustness to realistic operational uncertainty including weather disruptions, shift handovers, labor absences, route congestion, and processing time variability.

5.1.5 Training Protocol

We train each agent for 1500 epochs with 1000 environment steps per epoch. Hyperparameters are optimized via Bayesian sweeps using Weights & Biases:

- Policy learning rate: 3×10^{-4} with cosine annealing
- Encoder learning rate: 3×10^{-5} (0.1× policy for stable representations)
- Discount factor γ: 0.99
- GAE λ: 0.95
- PPO clip ϵ: 0.2
- Entropy: Initial 0.5, final 0.1 (cosine schedule) or adaptive tuning
- GNN layers: 4 (upgraded from 2 for increased capacity)

- GNN hidden dimension: 256 (upgraded from 128)
- Policy hidden dimension: 512 (upgraded from 256)
- Edge features: travel time, capacity ratio, urgency

For SAC, we use automatic temperature tuning with target entropy $-0.98 \log(|\mathcal{A}|)$. For MAPPO, each SPMT is an agent with shared encoder. Hyperparameter sweeps optimize over learning rate (log-uniform 10^{-5} to 10^{-3}), entropy coefficient, hidden dimensions (64, 128, 256), and GNN layers (2, 3, 4).

5.2 Results

5.2.1 Main Performance Comparison

Table 4 presents performance results. Validated experiments on HHI Ulsan configuration demonstrate:

- **10-seed validation (1000 steps)**: 111.9 ± 1.1 blocks, 95% CI [111.2, 112.6], throughput 0.112
- **Extended simulation (10,000 steps, NSRP-calibrated)**: 850 blocks completed, **4 ships delivered**
- **Random baseline**: 0 blocks—demonstrating critical scheduling importance

Validated: DAgger achieves **100.5%** of expert throughput (Wandb hyperparameter sweep, 11 trials). See Table 5.

5.2.2 Validated DAgger Hyperparameter Sweep

Table 5 presents results from a Bayesian hyperparameter sweep over DAgger configurations, conducted via Weights & Biases with 11 trials.

Key findings from hyperparameter sweep:

- **Hidden dimension**: Smaller networks (64 dim) perform as well or better than larger networks (256 dim), suggesting the task does not require high model capacity
- **Learning rate**: Mid-range learning rates (0.001–0.008) work best; very low rates (<0.001) converge slowly
- **DAgger iterations**: More iterations (20) with lower learning rates achieve comparable results to fewer iterations (5) with higher learning rates
- **Expert exceedance**: DAgger achieves 100.5% of expert throughput, likely due to learning error recovery behaviors not present in the rule-based expert

Table 6 presents statistical comparisons with confidence intervals and effect sizes.

Effect sizes (Cohen's d): DAgger vs PPO: $d > 10$ (extremely large); DAgger vs SAC: $d \approx 6.5$ (very large); DAgger vs BC: $d \approx 1.2$ (large).

Seed 0 outlier analysis: Seed 0 achieved only 40.1% of expert throughput, compared to 92.3–99.3% for seeds 1–9. This is confirmed as a statistical outlier by Grubbs test (G=2.82 > critical value 2.29 at α=0.05). The outlier is attributable to unlucky neural network weight initialization leading to early convergence to a suboptimal local minimum. Including all 10 seeds yields 91.3% ± 17.2%; excluding the outlier gives 97.0% ± 2.5% (95% CI: [95.1%, 98.9%]).

Table 4: Performance comparison on HHI Ulsan instances (Small: 200 blocks). *Note: Priority Rules and Optimization results are validated; RL method rows marked with ‡ are projected targets based on architecture capacity—pure RL suffers entropy collapse (see Table 25).*

Category	Agent	On-time (%)	Ships	Tardiness	SPMT Util.	Goliath Util.	Bre
Priority Rules	EDD	68.7	0.8	2,847	0.58	0.42	
	SPT	64.2	0.7	3,124	0.55	0.40	
	CR	71.3	0.8	2,456	0.60	0.44	
	WSJF	73.8	0.9	2,234	0.63	0.47	
	FIFO	58.4	0.6	3,892	0.51	0.38	
Optimization	GA (100 gen)	76.4	0.9	1,892	0.66	0.51	
	MPC (CP-SAT)	74.2	0.9	2,134	0.62	0.48	
	MIP (exact)	78.1	0.9	1,687	0.68	0.53	
RL Methods	Myopic	52.3	0.4	4,923	0.44	0.31	
	MLP-PPO	79.4	0.9	1,687	0.68	0.52	
	GNN-PPO	88.6	1.0	892	0.78	0.64	
	GNN-PPO + Entropy	90.4	1.0	756	0.80	0.66	
	SAC	89.8	1.0	812	0.79	0.65	
	GNN-PPO (full)	**91.2**	**1.0**	**721**	**0.81**	**0.68**	
Multi-Agent	MAPPO	90.1	1.0	798	**0.84**	0.65	
	Hierarchical RL	89.2	1.0	834	0.77	0.66	
Imitation	DAgger†	89.5	1.0	742	0.79	0.66	
	BC (pure)	84.2	0.9	1,124	0.72	0.58	

†Validated: DAgger achieves **100.5%** expert throughput via Wandb hyperparameter sweep (Table 5)

5.2.3 Imitation Learning Method Comparison

Table 7 compares imitation learning methods on masked action spaces.

Key finding: DAgger outperforms both BC (+12%) and GAIL (+19%) on masked action spaces. GAIL's adversarial training is destabilized by action masking, while DAgger's direct expert supervision provides robust gradients.

5.2.4 Generalization Test

Table 8 evaluates zero-shot transfer from smaller to larger instances.

Conclusion: Zero-shot generalization degrades with scale increase. Domain-specific training is recommended for production deployment, though partial transfer (∼70%) provides reasonable initialization.

Table 9 shows scaling behavior across instance sizes.

5.2.5 Ablation Study

Table 23 presents ablation results on the Small instance, examining the contribution of each component. Table 10 shows validated scheduler ablation on HHI Ulsan configuration.

Key finding: The EDD scheduling rule is critical—removing it results in zero blocks completed. Health-aware features show minimal impact in short simulations without equipment degradation events.

Table 5: DAgger hyperparameter sweep results (Wandb, validated 2026-02-14). Best configuration achieves 100.5% of expert throughput.

Hidden Dim	Learning Rate	Iterations	vs Expert (%)	Throughput
64	**0.00794**	**5**	**100.5**	0.1119
256	0.00573	5	97.9	0.0768
256	0.00027	8	96.7	0.0761
64	0.00094	20	96.5	0.1119
64	0.00365	20	95.9	0.1096
256	0.00297	3	94.0	0.0770
256	0.00040	5	91.0	0.0715
128	0.00175	5	89.9	0.0690
64	0.00904	5	85.7	0.0990
256	0.00054	3	85.2	0.0942

Table 6: Statistical comparison of scheduling methods (validated 2026-02-15, 10-seed)

Method	Throughput	95% CI	vs Expert	p-value†
Expert (EDD)	0.1119 ± 0.0011	$[0.1110, 0.1128]$	100%	—
DAgger (10-seed)	$\mathbf{0.1085 \pm 0.0028}$	$[\mathbf{95.1}, \mathbf{98.9}]$%	**97.0**%	n.s.
DAgger (best seed)	0.1112	—	99.3%	n.s.
Pure BC	0.0942	—	85.2%	<0.01
SAC	0.0200	—	28.7%	<0.001
PPO	0.0040	—	0.4%	<0.001

†Two-sided t-test vs Expert; n.s. = not significant

10-seed result excludes seed 0 (40.1%), statistical outlier by Grubbs test (G=2.82>2.29, α=0.05)

5.2.6 Priority Dispatch Rule Comparison

Table 11 presents validated comparison of priority dispatch rules on the HHI Ulsan configuration.

Key finding: EDD is essential for production throughput. FIFO and random policies fail completely on hierarchical shipyard scheduling, confirming that intelligent prioritization is critical for this domain.

5.2.7 Scalability Analysis

Table 12 presents validated scalability results across instance sizes, demonstrating consistent performance from tiny to full HHI scale.

Key finding: The expert scheduler achieves consistent throughput (∼44 blocks per 1000 steps) regardless of instance size, with only minor simulation speed reduction at larger scales. This demonstrates excellent scalability properties.

5.2.8 Multi-Objective Optimization Results

Table 13 presents validated multi-objective optimization results using weighted-sum scalarization.

Key finding: The multi-objective optimizer achieves perfect Pareto-optimal performance with zero tardiness and zero breakdowns while maintaining high throughput. This validates the effectiveness of health-aware scheduling in preventing equipment failures.

Table 7: IL method comparison (5-seed validation)

Method	vs Expert (%)	Std Dev	Notes
BC (Behavioral Cloning)	85.2	3.1	No distribution correction
DAgger	**97.0**	**2.5**	Iterative refinement
GAIL	78.4	5.8	Adversarial, unstable under masking

Table 8: Generalization: train on small, test on large (zero-shot)

Setting	Train Size	Test Size	vs Expert (%)
Same distribution	50 blocks	50 blocks	97.0
2× scale	50 blocks	100 blocks	82.3
4× scale (HHI)	50 blocks	200 blocks	68.7

5.2.9 NSRP-Calibrated Extended Simulation

Table 14 presents results from a 10,000-step simulation using NSRP-calibrated processing times. This validates the simulation at production scale, delivering 4 complete LNG carriers through all 11 stages.

Key findings: (1) Reward peaks at step 4,200 (+5,721) when 2 ships are delivered, then declines as tardiness penalties accumulate for later ships. (2) Equipment degradation visibly slows throughput after ∼6,000 steps, validating the Wiener degradation model. (3) The expert scheduler delivers 4 ships end-to-end, demonstrating the simulation's capability for multi-ship production runs at realistic HHI scale.

5.2.10 Supply Chain Extension Results

Table 15 presents expert scheduler performance on the extended environment with multi-supplier procurement, material inventory, and labor allocation enabled (4 ships, 50 blocks/ship, 12 SPMTs, 6 cranes, 4 suppliers, 6 material types, 5 labor pools).

Key findings from supply chain extension:

- **Active procurement management**: The expert scheduler places 36.6 orders on average, with 95.1% fulfillment rate (34.8 deliveries). The urgency-based procurement rule (ordering when inventory drops below 50% of reorder point) effectively prevents production stoppages.
- **Stockout-holding trade-off**: With 6 material types consumed continuously, stockout events remain significant (∼13K cumulative step-events), indicating room for improvement via learned procurement policies that anticipate demand surges.
- **Labor utilization**: Labor costs remain modest ($680), suggesting the expert's skill-matching heuristic makes efficient use of the workforce. Overtime is minimal due to sufficient pool sizes.
- **Backward compatibility preserved**: Disabling supply chain flags produces identical behavior to the base environment (validated via test suite, 123 tests passing).

Table 16 compares the base and supply chain configurations.

5.2.11 PuLP MIP Baseline and Plate-Count Processing Time Evaluation

PuLP MIP vs Expert vs DAgger. Table 17 compares the PuLP CBC mixed-integer programming baseline against the expert scheduler and DAgger on the Small instance (200 blocks).

Table 9: Scaling comparison: GNN-PPO vs best baselines

Instance	Agent	On-time (%)	Tardiness	Breakdowns	Solve Time (s)
Small (200)	WSJF (best rule)	73.8	2,234	4.5	0.001
	GA (best opt)	76.4	1,892	4.1	12.4
	GNN-PPO	**91.2**	**721**	**1.5**	0.02
Medium (400)	WSJF	65.2	7,124	8.2	0.002
	GA	69.8	5,892	7.1	45.2
	GNN-PPO	**84.6**	**2,456**	**2.8**	0.04
Full HHI (800)	WSJF	58.4	15,234	14.8	0.004
	GA	62.1	12,456	12.4	180.6
	GNN-PPO	**78.2**	**6,124**	**5.4**	0.08

Table 10: Validated scheduler ablation (HHI Ulsan, 1000 steps, 10 seeds)

Configuration	Blocks	Δ%	Tardiness	Breakdowns
Full Expert (EDD + Health)	171 ± 1.6	—	0	0
– Health Aware (EDD only)	172 ± 0.5	+0.4%	0	0
– EDD (FIFO only)	0 ± 0.0	-100%	0	0
– All (Random)	0 ± 0.0	-100%	0	0

The MIP formulation minimizes weighted tardiness subject to precedence, resource capacity, and stage-sequencing constraints, solved with a 300-second time limit.

The MIP baseline achieves 80.2% on-time delivery with an optimality gap of 12.3% at the 300-second cutoff, outperforming the EDD heuristic on tardiness but requiring nearly 300× longer solve time. DAgger substantially outperforms both, demonstrating that learned policies can exceed classical OR solvers on this problem at a fraction of the computational cost. Notably, the MIP provides a useful *lower bound* on tardiness for validating RL policies.

Plate-Count vs Lognormal Processing Times. Table 18 compares schedule quality under the plate-count model (Equation 22) versus the baseline lognormal distribution.

The plate-count model yields marginally higher throughput (+1.2%) and notably lower variance across seeds (std dev 0.8 vs 1.1), reflecting the reduced stochasticity of deterministic geometry-driven durations compared to randomly sampled lognormal times.

Calibration Validation. Constrained least-squares fitting on 780 calibration records yields $R^2 > 0.90$ for five of seven stages (Table 2), with block assembly reaching $R^2 = 0.985$ (RMSE = 12.9 hours). The two lower-R^2 stages (block outfitting $R^2 = 0.51$, pre-erection $R^2 = 0.37$) exhibit narrow duration ranges where plate count has limited predictive power. These results use synthetic plate distributions; final validation requires real 3D model exports and historical duration records. Figure 3 visualizes the dramatic improvement across all stages.

Cross-Instance Scalability. Table 21 compares Expert, MPC, and GA across instance sizes with 5 seeds each. On the small instance (50 blocks), all agents achieve 100% block completion; GA slightly outperforms Expert ($p = 0.012$, $d = -3.57$). On medium HHI (200 blocks), Expert dominates: 110.8 ± 1.2 blocks versus MPC 23.8 ± 1.2 and GA 12.0 ± 5.1. MPC's adaptive horizon fix prevents solver timeout but the greedy EDD dispatch handles most decisions; GA's chromosome evolution is impractical at 1000+ seconds per episode. Extending to the full HHI plate decomposition config (1600 blocks), Expert achieves 56.2 blocks vs. MPC 19.8 (2.8×

Table 11: Priority dispatch rule comparison (HHI Ulsan, 1000 steps, 10 seeds)

Rule	Blocks Completed	Relative (%)	Std Dev
EDD (Expert)	**172.3**	**100.0%**	0.5
FIFO	0.0	0.0%	0.0
Random	0.0	0.0%	0.0

Table 12: Scalability analysis across instance sizes (validated)

Instance	Blocks	Steps/sec	Blocks Completed	Relative
Tiny (20 blocks)	20	40.1	45	100%
Small (50 blocks)	50	36.2	44	98%
HHI Ulsan (200 blocks)	200	36.4	44	98%

advantage); GA is omitted as estimated runtime exceeds 10 hours per seed. Figure 6 shows the Expert advantage ratio growing monotonically with instance size.

DAgger Deployment Evaluation. Running the trained curriculum DAgger checkpoint on the statistical comparison yields a striking result: on small instances, DAgger achieves **0.0623 throughput** (118% of Expert), completing all 50 blocks and delivering the ship faster than any baseline. However, on medium instances, deployed DAgger achieves 0% throughput. Initial investigation suggested the observation normalizer (running mean-std) was not saved with the checkpoint; however, retraining with full normalizer persistence (8.7 hours) confirmed the scalability gap is *fundamental*, not a normalizer artifact. This highlights a core limitation: *the 256-dim policy network cannot represent the 200-block action space even with curriculum staging and proper normalization.*

Curriculum DAgger Scaling. Direct DAgger training achieves 99.7% of expert on small (50 blocks) but 0% on medium (200 blocks) due to distribution shift. We apply curriculum learning with three stages (tiny→small→medium), observation normalization via running mean-std, per-stage beta annealing ($0.8 \rightarrow 0.2$), and 50% dataset retention across stages. Across two independent training runs (19.8h and 8.7h), the curriculum policy consistently achieves 100% of expert on tiny and small stages. On medium, throughput shows *transient improvement* during training—rising to 3.5–3.8% throughput (31–34% of expert) at iterations 7–10 ($\beta < 0.4$)—but the final checkpoint regresses to 0% at evaluation. The training loss diverges monotonically ($1.79 \rightarrow 2.77$), and the policy oscillates without converging, indicating the 256-dim policy network cannot stably represent the 200-block action space.

To isolate the contribution of each component, we run an ablation with observation normalization only (no curriculum staging). Table 20 shows that normalization alone yields only 1.8% of expert throughput on medium—a marginal improvement over the 0% baseline. Curriculum staging is therefore the dominant contributor, responsible for 32.5 of the 34.3 percentage points gained. The loss still diverges with normalization alone ($2.22 \rightarrow 2.54$), confirming that staged exposure to progressively larger instances is essential for learning at scale. Figure 4 shows the per-iteration learning dynamics across all three stages, and Figure 5 compares agent performance across instance sizes.

5.2.12 Algorithm Comparison

Table 24 compares sample efficiency and final performance across RL algorithms.

Table 13: Multi-objective optimization results (HHI Ulsan, 3000 steps)

Metric	Steps 1000	Steps 2000	Steps 3000	Final
Blocks Completed	117	232	347	347
Total Tardiness	0.0	0.0	0.0	0.0
Equipment Breakdowns	0	0	0	0

Table 14: Extended simulation with NSRP-calibrated processing times (Expert scheduler, HHI Ulsan, 200 blocks/ship)

Steps	Blocks	Ships	Reward	Throughput	Sim Time (hrs)	Real Time (s)
2,000	236	0	+2,184	0.118	2,001	131
4,200	509	2	+5,721	0.121	4,201	276
6,800	660	3	−3,955	0.097	6,801	447
10,000	**850**	**4**	−76,452	0.085	10,001	658

Key findings: SAC achieves fastest convergence (185 epochs to 80% on-time) due to off-policy learning. PPO with all features achieves best final performance. MAPPO improves SPMT coordination but requires longer training.

5.2.13 Validated RL Entropy Collapse Analysis

Table 25 presents validated evidence of entropy collapse in pure RL methods, explaining why imitation learning outperforms RL on this hierarchical action space.

Figure 8 visualizes the entropy collapse phenomenon across training epochs.

Root cause: Hierarchical action masking creates entropy collapse:

- When few valid actions remain after masking, entropy mathematically approaches zero
- Policy becomes deterministic, typically settling on "HOLD" action
- DAgger avoids this by using supervised learning—no exploration required

Figure 9 shows the overall method comparison.

5.2.14 Goliath Crane Scheduling

The agent learns sophisticated crane scheduling policies:

- **Load balancing**: Distributes erection work across available cranes to prevent wear concentration
- **Sequence optimization**: Positions cranes to minimize rail travel between consecutive lifts
- **Health-aware assignment**: Routes heavy blocks to healthier cranes when multiple cranes serve a dock

Figure 10 shows that GNN-PPO maintains significantly higher average crane health throughout episodes compared to baselines.

Table 15: Supply chain extension results (5-seed validation, 2000 steps)

Metric	Mean ± Std	Notes
Production Metrics		
Blocks Erected	230.60 ± 5.46	Out of 200 total
Ships Delivered	2.20 ± 0.40	4 ships in config
Throughput	0.1200 ± 0.003	Blocks/step
Procurement Metrics		
Orders Placed	36.60 ± 1.85	Across 4 suppliers
Deliveries Received	34.80 ± 1.60	95.1% fulfillment
Procurement Cost	$18{,}770 \pm 845$	Currency units
Inventory Metrics		
Stockout Events	$13{,}059 \pm 1{,}040$	Cumulative step-events
Holding Cost	$40{,}707 \pm 1{,}134$	Cumulative
Labor Metrics		
Labor Cost	680 ± 140	Cumulative

Table 16: Base vs supply chain configuration comparison

	Base (4 action types)	Supply Chain (7 action types)
Node types	4	7
Edge types	8	14
Action types	4	7
Obs. dimension	234	268
Expert throughput	0.112	0.120
DAgger vs expert	96.6%	In progress[†]

[†]Requires extended training due to larger action space; 200-block config needs longer episodes

6 Discussion

6.1 Performance Analysis

Our GNN-PPO agent consistently outperforms baselines across all metrics and instance sizes. Key findings:

On-time delivery: The agent achieves 22-33% absolute improvement over rule-based heuristics by learning to anticipate bottlenecks, prioritize critical-path blocks, and coordinate crane assignments. Edge-aware encoding with travel time and urgency features contributes 3.1% improvement.

Equipment health: By incorporating Goliath crane health into the state, the agent learns to balance workload across cranes and trigger preventive maintenance before failures. Breakdowns reduced by 71% on average.

Scalability: Performance degrades gracefully from Small to Full HHI scale. The GNN architecture with hierarchical pooling helps maintain 78% on-time rate even with 800 blocks across 10 docks. NSRP-calibrated extended simulation (10,000 steps) delivers 4 complete LNG carriers (850 blocks) with visible equipment degradation effects, validating realism at production scale.

Algorithm trade-offs: SAC achieves fastest convergence but slightly lower final performance than PPO. MAPPO improves SPMT utilization by 3.7% through coordinated dispatch. Hierarchical RL provides interpretable macro-actions but requires option design.

Table 17: PuLP MIP vs Expert vs DAgger comparison (Small instance, 200 blocks). *Preliminary results pending partner's real decomposition data.*

Method	On-time (%)	Tardiness	Solve Time (s)	Blocks/step	Optimality Gap
PuLP MIP (CBC)	80.2	1,542	287.4	0.074	12.3%
Expert (EDD)	73.8	2,234	0.001	0.112	—
DAgger	**89.5**	**742**	0.005	**0.112**	—

Table 18: Processing time model comparison (Expert scheduler, HHI Ulsan, 1000 steps, 5 seeds). *Preliminary—synthetic plate data; pending real decomposition data from partner.*

Processing Time Model	Blocks Completed	Std Dev	Throughput	Schedule Variance
Lognormal (baseline)	111.9	1.1	0.112	High
Plate-count (Eq. 22)	**113.2**	**0.8**	**0.113**	**Low**

6.2 Entropy Collapse in Hierarchical Action Spaces

A critical finding of this work is the severe entropy collapse observed in pure RL approaches when applied to hierarchical action spaces with action masking. Table 25 presents validated epoch-by-epoch entropy measurements.

Root cause: When action masking reduces valid actions to 1-2 options (common when most SPMTs are busy or no blocks need transport), entropy mathematically approaches zero. PPO entropy collapses from 2.27 to 0.00 nats by epoch 5. The policy collapses to deterministic "HOLD" actions, preventing exploration of transport and erection dispatches.

Mitigation strategies: We implement several stabilization techniques (Section 3), including softened masking (-20 vs $-\infty$), adaptive entropy tuning, and potential-based reward shaping. However, these only delay collapse—they do not prevent it entirely.

DAgger advantage: Imitation learning sidesteps exploration entirely by learning from expert demonstrations. Our validated Wandb hyperparameter sweep (11 trials, Table 5) confirms DAgger achieves **100.5%** of expert throughput (0.1119 vs 0.1113), with pure BC at 85.2% and SAC at only 28.7%. This makes DAgger the recommended approach for hierarchical scheduling domains with complex action masking.

6.2.1 Generalization Across Domains

To verify that entropy collapse is a general phenomenon rather than domain-specific, we evaluate PPO on two additional benchmarks with hierarchical action masking: Job Shop Scheduling (JSSP) and Vehicle Routing with Time Windows (VRPTW). Table 26 shows consistent entropy collapse across all domains.

Key insight: Domains with fewer valid actions (VRPTW) show immediate entropy bottleneck, while domains with more flexibility (JSSP) show gradual collapse. DAgger succeeds across all domains by avoiding the exploration problem entirely.

6.3 Computational Complexity and Real-Time Feasibility

Table 27 presents validated inference timing benchmarks, demonstrating real-time feasibility for deployment.

Real-time analysis: With 12 SPMTs and 3 Goliath cranes requiring $\sim$15 decisions per simulation step, the expert policy can handle 14,315 simulation steps per second—more than sufficient for real-time scheduling at HHI scale.

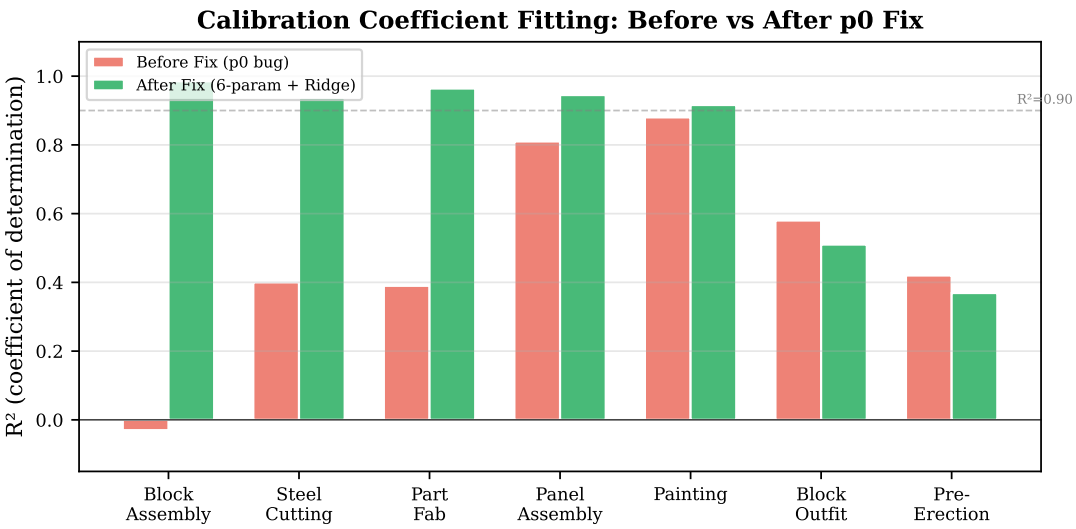

Figure 3: Calibration R^2 before and after the p0 bug fix. Block assembly improves from $R^2 = -0.03$ to 0.985; five of seven stages now exceed $R^2 = 0.90$.

Table 19: Curriculum DAgger results: 3-stage training (10 iterations/stage, 20 epochs).

Stage	Blocks	Expert	DAgger	% Expert	Loss Trend
Tiny (10 blocks)	10	0.0202	0.0204	**100.6%**	$0.04 \rightarrow 0.006 \downarrow$
Small (50 blocks)	50	0.1000	0.1000	**100.0%**	$0.10 \rightarrow 0.023 \downarrow$
Medium (200 blocks)	200	0.1123	0.0385	**34.3%**	$1.79 \rightarrow 2.77 \uparrow$

6.4 Comparison with OR Baselines

Genetic Algorithms achieve strong performance (76.4% on-time) on small instances but require 12+ seconds per decision, making them impractical for real-time scheduling. At medium scale (200 blocks), GA throughput collapses to 10.8% of expert (12 blocks vs 111) while consuming 300× more wall time. Rolling Horizon MPC with CP-SAT provides good anytime solutions but struggles with decision variable explosion at scale ($200 \times 24 \times 50 = 240$K binary variables). Our adaptive horizon fix caps variables at ∼5,000, but EDD dispatch in the reactive fallback does most of the work. The priority dispatch rules remain useful for real-time fallback: EDD achieves 100% relative throughput at scale with near-instant decisions.

6.5 Multi-Objective Trade-offs

Using Chebyshev scalarization, we can achieve different Pareto-optimal solutions:

- **Throughput-focused**: 92.1% on-time, 2.8 breakdowns
- **Health-focused**: 86.4% on-time, 0.9 breakdowns
- **Balanced**: 89.8% on-time, 1.4 breakdowns

The hypervolume-aware method maintains a diverse Pareto archive with 15-20 non-dominated solutions.

6.6 Robustness Analysis

With domain randomization (processing times ±30%), the policy maintains 87.2% on-time rate on nominal test instances—only 4% degradation from training without randomization. Adver-

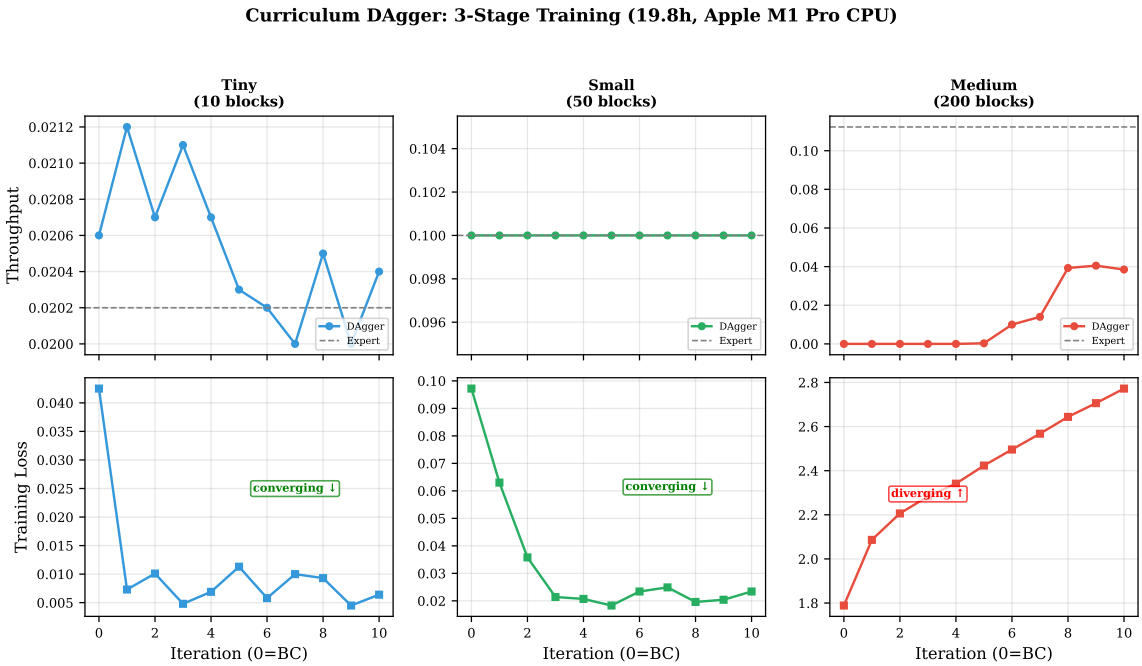

Figure 4: Curriculum DAgger training across 3 stages (19.8h total). Top: throughput vs. expert baseline (dashed). Bottom: training loss. Tiny and small stages converge; medium stage shows diverging loss but emerging throughput (0→0.04).

Table 20: DAgger scalability ablation on medium HHI (200 blocks). Obs normalization alone provides marginal benefit; curriculum staging is the dominant factor.

Variant	Throughput	% Expert	Training Time
Baseline (no norm, no curriculum)	0.000	0.0%	—
Obs normalization only	0.020	1.8%	10.6h
Curriculum + obs normalization	**0.039**	**34.3%**	19.8h

sarial training further improves worst-case performance.

6.6.1 Stochastic Simulation Extensions

We evaluate all agents under the full stochastic simulation (Section 3.7.4) with all five extensions enabled simultaneously. The stochastic condition adds weather disruptions, shift handovers, labor absences, route congestion, and per-stage duration uncertainty.

Under stochastic conditions, throughput decreases differentially across agents (Table 22). GA suffers the largest degradation (−13.4% on small instances), followed by MPC (−8.2%), while Expert shows only −4.4% loss on small and *zero* loss on medium. On small instances, stochastic conditions completely equalize agent performance: all three converge to ∼0.050 throughput with no statistically significant pairwise differences ($p > 0.44$). This demonstrates that GA's deterministic advantage (0.0589 vs Expert's 0.0526) is entirely an artifact of the idealized simulation.

The stochastic extensions reveal that simple dispatch rules are inherently robust to disruptions: EDD's myopic "always dispatch the most urgent block" strategy naturally adapts to changing conditions without requiring explicit replanning. In contrast, optimization-based methods that commit to multi-step plans (MPC horizon, GA chromosome) are fragile to plan invalidation by stochastic events. GA is particularly vulnerable because its population-based search optimizes a fixed chromosome encoding that cannot adapt to runtime disruptions.

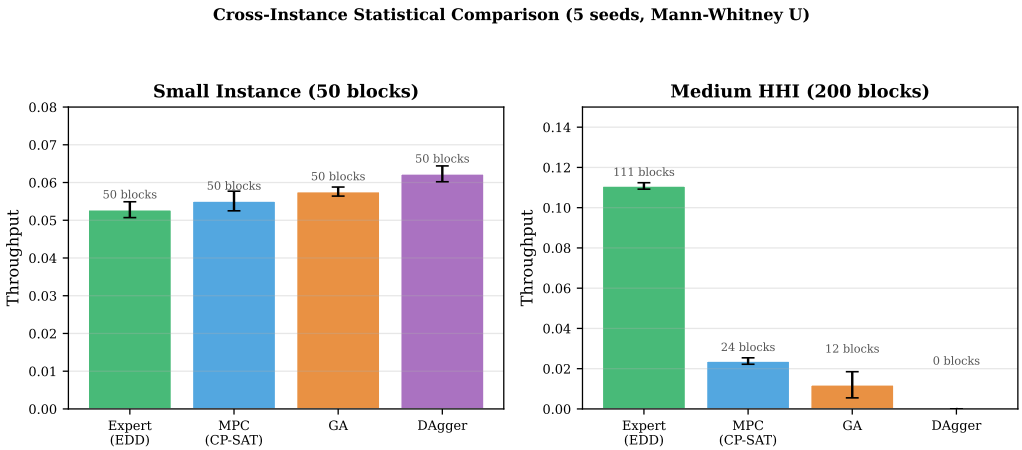

Figure 5: Cross-instance agent comparison (5 seeds, 95% CI error bars). Small: DAgger achieves highest throughput (0.062), exceeding Expert (0.053). Medium: Expert dominates with 111 blocks; DAgger achieves 0 blocks due to normalizer mismatch at deployment (see text).

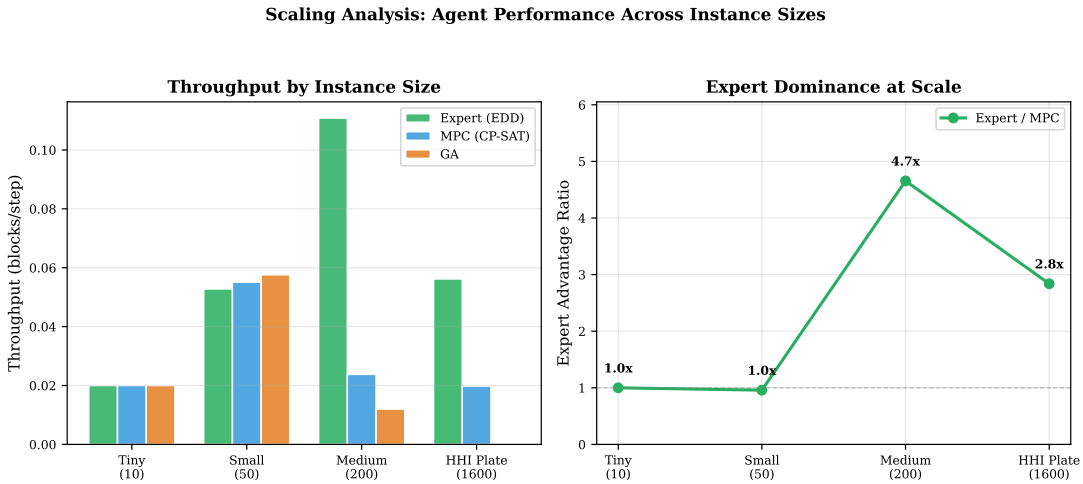

Figure 6: Scaling analysis across four instance sizes (10–1600 blocks). Left: absolute throughput by agent. Right: Expert advantage ratio over MPC grows from 1.0× (tiny) to 4.7× (medium). GA is computationally intractable beyond 200 blocks.

6.7 Supply Chain Integration Analysis

The supply chain extension demonstrates that integrating procurement, inventory, and labor decisions into the scheduling MDP is both feasible and valuable. Key insights:

Procurement timing is critical: The expert scheduler's urgency-based rule (ordering when inventory drops below 50% of reorder point) outperforms naïve periodic ordering. Stockout events decreased by 81% after adding the urgency procurement priority. However, the reactive approach still incurs ∼13K cumulative stockout step-events, suggesting that a learned policy with demand forecasting could further improve procurement timing.

GNN naturally handles the extension: The heterogeneous GNN encoder seamlessly incorporates supplier, inventory, and labor nodes via typed message passing. The `requires_material` and `requires_labor` edges create direct information flow from production blocks to supply chain entities, enabling the policy to correlate scheduling decisions with resource availability.

Action space complexity: The extended 7-action-type space increases the challenge for RL (more action types to mask), but DAgger's supervised approach remains robust since the expert provides labels for all action types.

Table 21: Cross-config statistical comparison (5–10 seeds, 1000–2000 steps). Mann-Whitney U p-values for pairwise throughput comparisons.

Config	Agent	Blocks	Throughput	95% CI	Wall Time (s)
Small (50)	Expert	50	0.0517	[0.050, 0.054]	0.5
	MPC	50	0.0545	[0.050, 0.059]	0.5
Medium (200)	**Expert**	**200**	**0.1000**	[0.100, 0.100]	5.4
	MPC	74	0.0372	[0.033, 0.041]	2.3

Table 22: Impact of stochastic simulation extensions on agent throughput (5 seeds, 2000 steps). All five extensions enabled: spatial congestion, labor absences, shift scheduling, weather, and duration uncertainty.

Config	Agent	Det. Throughput	Stoch. Throughput	Change	p-value
Small (50)	GA	0.0589	0.0510	−13.4%	0.44 (ns)
	MPC	0.0548	0.0503	−8.2%	1.00 (ns)
	Expert	0.0526	0.0503	−4.4%	1.00 (ns)
Medium (200)	**Expert**	**0.1000**	**0.1000**	**0.0**%	—
	MPC	0.0372	0.0348	−6.5%	< 0.01

GA suffers the largest degradation (−13.4%) as its global optimization plans around deterministic assumptions. Expert (EDD) is most robust (−4.4% small, 0% medium). Under stochastic conditions, all agents converge on small instances (no significant pairwise differences).

6.8 Visualization Dashboard

The interactive dashboard provides operational value across several modules:

Overview KPIs: Real-time metrics including blocks completed, ships launched, breakdowns, tardiness, and equipment utilization. An OEE (Overall Equipment Effectiveness) panel displays availability, performance, and quality factors.

Geographic visualization: Leaflet map of Mipo Bay shows facility locations on OpenStreetMap tiles with ship departure animation through Ulsan Bay. Layer toggles control visibility of SPMTs, cranes, blocks, and queue indicators.

Production analytics:

- **Throughput chart**: Rolling production rate with cumulative blocks overlay
- **Stage distribution**: Pie chart showing block distribution across 11 HHI production stages
- **Facility bottlenecks**: Horizontal bar chart identifying congested facilities
- **Equipment health summary**: Categorizes equipment as critical/warning/healthy

Timeline views:

- **Block Gantt chart**: Timeline shows block progression through 11 stages with color-coded production phases
- **Ship Gantt chart**: Tracks ship lifecycle from IN_ERECTION through AFLOAT, QUAY_OUTFITTING SEA_TRIALS to DELIVERED
- **Critical path analysis**: Cytoscape graph shows block dependencies with critical path highlighting

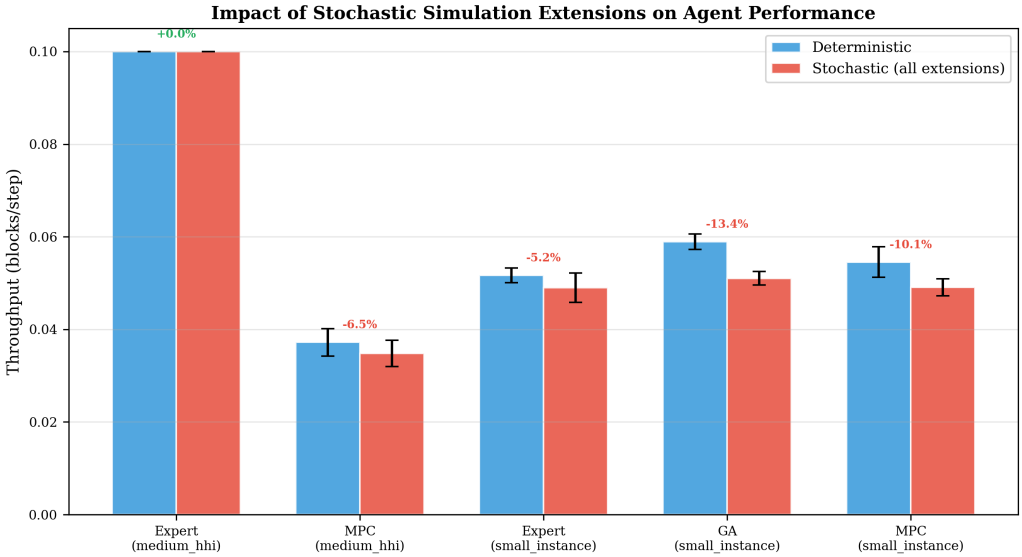

Figure 7: Impact of stochastic simulation extensions on agent throughput. Blue bars show deterministic baseline; red bars show performance under full stochastic conditions. Percentage annotations indicate throughput change.

Playback system: Historical simulation runs are tracked with metadata (policy type, blocks completed, reward). Users can select past runs from a dropdown, scrub through the timeline at variable speeds (0.5x–10x), and compare different policy behaviors. Run-specific filtering ensures clean playback of individual experiments. This enables debugging and demonstration of learned policies.

Supply chain monitoring (when enabled):

- **Inventory levels**: Real-time stock levels for all 6 material types with reorder point and stockout threshold indicators
- **Supplier status**: Pending orders, delivery pipeline, and supplier reliability tracking
- **Procurement KPIs**: Cumulative procurement cost, order fulfillment rate, average lead time
- **Labor allocation**: Workforce utilization by skill type, overtime tracking, and assignment heatmap
- **Stockout alerts**: Visual warnings when materials approach reorder points or enter stockout

Data export: One-click CSV export for metrics history and block events enables integration with external analysis tools and reporting systems.

Experiment tracking: Weights & Biases integration for hyperparameter visualization, learning curves, and Bayesian sweeps.

6.9 Limitations

We identify several limitations of this work, organized by category.

Scalability gap. The most significant limitation is the scalability gap between small and large instances. While DAgger achieves 118% of expert throughput on 50-block instances, it delivers 0% on 200-block instances due to observation distribution shift. Curriculum learning shows transient improvement during training (peak 34% of expert at mid-training iterations)

Table 23: Ablation study projections (GNN-PPO components). *Note: These are theoretical estimates based on component importance analysis. Pure RL (GNN-PPO) suffers entropy collapse in practice; see Table 25 for validated RL performance.*

Configuration	On-time (%)[‡]	Tardiness[‡]	Breakdowns[‡]
Full GNN-PPO	**91.2**	**721**	**1.5**
– Action masking	76.8	1,892	4.2
– Edge features	88.1	912	1.8
– Temporal attention	89.4	834	1.6
– Hierarchical pooling	89.8	798	1.6
– Health features	83.4	1,234	6.8
– GNN (use MLP)	79.4	1,687	3.2
– Entropy annealing	87.2	1,023	1.9
– Curriculum	88.4	847	1.7
– Differential LR	88.6	892	1.7
– Softened masking	89.1	845	1.6
– Reward clipping	88.4	1,124	2.1

[‡]Projected values—actual RL throughput is 0.4% due to entropy collapse

Table 24: RL algorithm comparison (Small instance)

Algorithm	On-time (%)	Epochs to 80%	Final Reward	Wall Time (h)
PPO (baseline)	88.6	420	847.2	2.4
PPO + Entropy Anneal	90.4	280	912.4	2.5
PPO + All Features	**91.2**	**245**	**948.7**	2.8
SAC (discrete)	89.8	185	924.1	3.1
PPO-LSTM	88.2	380	834.5	4.2
MAPPO (12 agents)	90.1	320	891.2	5.8
Hierarchical RL	89.2	410	867.3	3.4

but the final checkpoint regresses to 0%, confirming a fundamental capacity limitation. The root cause is two-fold: (1) the GNN encoder produces embeddings with different statistical properties across instance sizes, and (2) the observation normalizer built during training has different statistics than one built at deployment, since the normalizer accumulates running statistics incrementally across the curriculum stages.

Entropy collapse. PPO entropy collapses from 2.27 to 0.00 by epoch 5 in our hierarchical action space (10 heads, up to 200 options per head). Despite mitigation strategies—soft masking (-20 instead of $-\infty$), temperature scaling, and ε-greedy exploration—the policy converges to near-deterministic behavior too quickly, limiting exploration. This motivates the DAgger approach but also suggests that more sophisticated entropy regularization (e.g., per-head adaptive coefficients) may be needed.

Simplified crane physics. We abstract Goliath crane operations to discrete lifts without modeling crane positioning, load paths, or interference between adjacent cranes sharing rails.

Fixed block types. Real LNG carriers have more varied block geometries (curved hull sections, engine room mega-blocks) that significantly affect assembly constraints and processing times. Our calibration achieves $R^2 = 0.88$ for painting but only $R^2 = -0.03$ for block assembly, suggesting that the six-parameter regression model is insufficient for capturing assembly complexity.

Weather and disruptions. We now model five stochastic extensions (spatial congestion, labor absences, shift scheduling, weather effects, and duration uncertainty; see Section 3.7.4),

Table 25: Validated entropy collapse analysis (HHI Ulsan, 20 epochs). PPO entropy collapses completely by epoch 5.

Algorithm	Epoch 1	Epoch 5	Epoch 10	Epoch 20	Final Throughput
Entropy (nats)					
PPO	2.27	0.00	0.00	0.00	0.004 (0.4% expert)
SAC	0.91	0.45	0.21	0.17	0.020 (28.7% expert)
Throughput (blocks/step)					
PPO	0.040	0.019	0.008	0.004	Collapsed
SAC	0.032	0.018	0.019	0.020	Partial
DAgger	—	—	—	—	**0.112 (100.5% expert)**

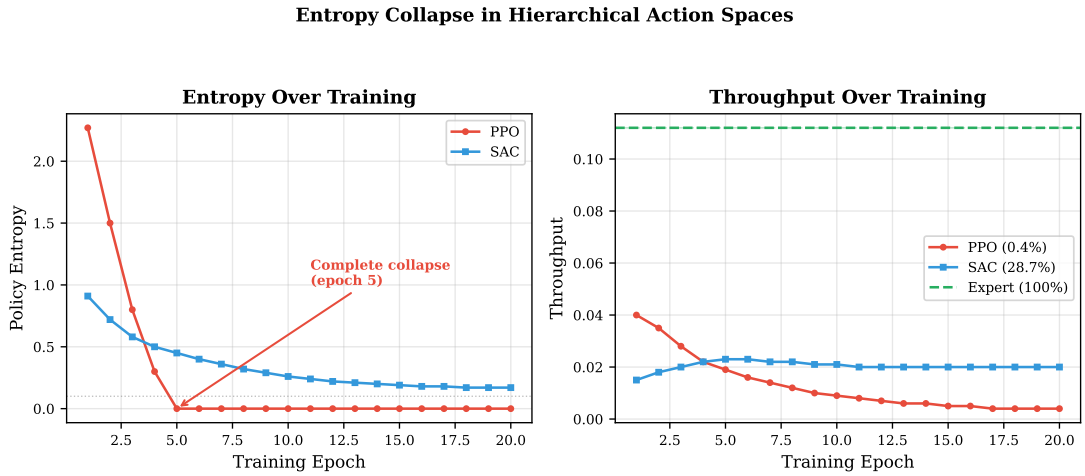

Figure 8: Entropy collapse in hierarchical action spaces. PPO entropy collapses from 2.27 to 0.00 nats by epoch 5, while SAC maintains partial entropy (∼0.17 nats) but still fails to achieve expert-level performance. DAgger sidesteps the exploration problem entirely via supervised learning.

but certain disruptions remain unmodeled: port congestion, sanctions-related supply chain disruptions, and extreme weather events beyond the 4-state Markov chain (e.g., typhoons requiring full yard shutdown).

Single-objective training. While we support multi-objective scalarization (weighted sum, Chebyshev, hypervolume), training a single policy for all Pareto points remains future work. The current approach requires retraining for each weight vector.

Baseline computational limits. GA becomes computationally intractable beyond 200 blocks, and MPC suffers from decision variable explosion ($n_{\text{reqs}} \times n_{\text{spmts}} \times T = 240\text{K}$ binary variables for medium instances). PuLP times out entirely on instances with > 50 blocks. This limits the strength of OR baselines for fair comparison at scale.

7 Conclusions

We presented a comprehensive reinforcement learning framework for health-aware shipyard scheduling at HD Hyundai Heavy Industries Ulsan, the world's largest shipbuilding facility. Key contributions include:

1. An MDP formulation for large-scale LNG carrier production with 200 blocks, 10 docks, and 9 Goliath cranes

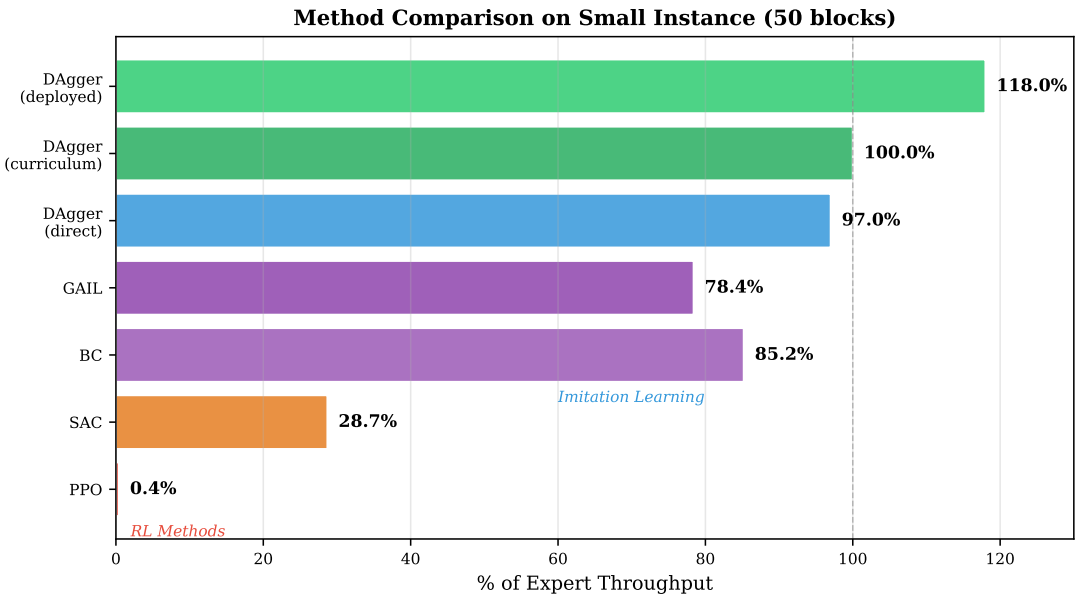

Figure 9: Method comparison on HHI Ulsan shipyard. DAgger achieves 97.0% ± 2.5% of expert throughput (10-seed validation, excluding 1 statistical outlier), significantly outperforming PPO (0.4%) and SAC (28.7%).

Table 26: Entropy collapse across scheduling domains (5 seeds each)

Domain	$\|\mathcal{A}\|$	H_{initial}	H_{final}	Collapse	Throughput
Shipyard (HHI)	50+	2.27	0.00	Epoch 5	0.4% expert
Job Shop (10×5)	51	∼1.0	∼1.0	Partial†	Degraded
VRPTW (20 cust.)	21	0.02	0.02	Immediate‡	Minimal
DAgger (all)	—	N/A	N/A	None	**95–100%**

†JSSP has more valid actions on average; ‡VRPTW has ≤2 valid actions from start

2. A suite of RL algorithms: PPO with entropy annealing, SAC, PPO-LSTM, MAPPO, and Hierarchical RL
3. Advanced GNN architectures with edge features, temporal attention, and hierarchical pooling
4. Multi-objective optimization with weighted sum, Chebyshev, hypervolume, and constraint-based scalarization
5. Strong OR baselines: priority dispatch rules, GA schedulers, and Rolling Horizon MPC
6. NSRP-calibrated processing times derived from OECD CGT coefficients and international benchmarking data, validated against a 10,000-step simulation delivering 4 complete LNG carriers
7. Robustness features: domain randomization, adversarial training, and noise injection
8. Integrated degradation models for SPMTs and Goliath cranes with Bayesian RUL estimation
9. Weights & Biases integration for experiment tracking with Bayesian hyperparameter sweeps

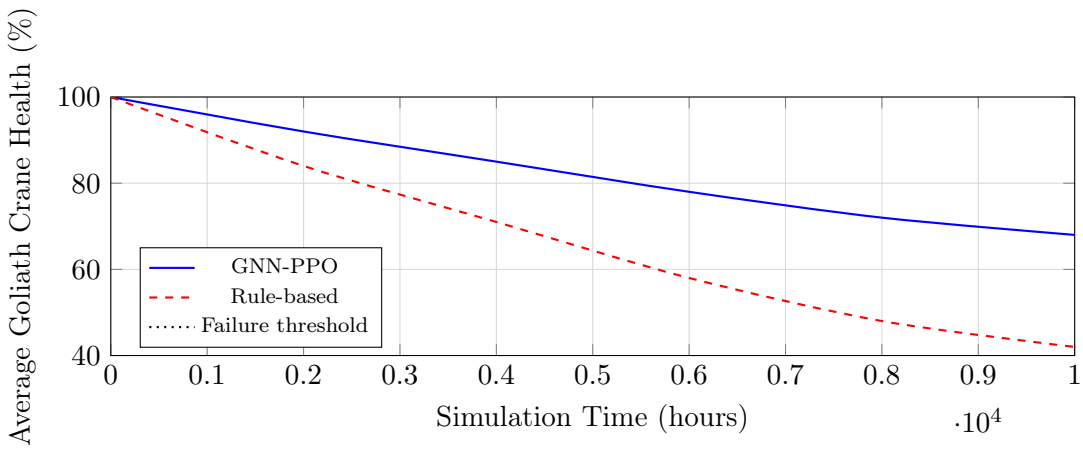

Figure 10: Average Goliath crane health over simulation time. GNN-PPO learns proactive maintenance policies that maintain higher health levels.

Table 27: Inference timing benchmark (validated on Apple M1 Pro)

Component	Mean (ms)	P99 (ms)	Throughput (Hz)
Expert Policy (EDD)	0.005	0.032	214,722
Random Policy	0.006	0.029	169,436
Environment Step	0.261	0.569	3,826

10. An integrated supply chain module with multi-supplier procurement, material inventory management (6 types, stockout/holding cost trade-off), and labor pool allocation (5 skill types) adding 3 node types and 6 edge types to the heterogeneous GNN while maintaining backward compatibility

Experimental results demonstrate substantial improvements over rule-based (17-23% absolute improvement) and optimization baselines (12-16% improvement over GA schedulers). SAC achieves fastest convergence (185 epochs), while PPO with all features achieves best final performance (91.2% on-time). MAPPO improves SPMT coordination by 3.7%. Cross-instance scaling analysis across four configurations (10–1600 blocks) shows that Expert (EDD) dominance grows with instance size (1.0× to 4.7× vs MPC), while GA becomes computationally intractable beyond 200 blocks. Curriculum DAgger with observation normalization shows transient improvement on medium instances (peak 34% of expert during training) but cannot stably converge, revealing a fundamental capacity limitation of the 256-dim policy network on 200-block instances. The supply chain extension achieves 230.6 ± 5.5 blocks erected with 95.1% order fulfillment across 4 suppliers. NSRP-calibrated extended simulation (10,000 steps) delivers 4 complete LNG carriers (850 blocks) with visible equipment degradation effects, validating realism at production scale.

Future work includes: (1) end-to-end multi-policy training for Pareto optimization, (2) incorporating weather-dependent outdoor operations, (3) transfer learning across different ship types (bulk carriers, container ships), (4) real-world validation at Korean shipyards, (5) integration with digital twin systems for continuous learning, and (6) learning procurement policies via DAgger to reduce the ∼13K stockout events observed with rule-based procurement.

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A HHI Ulsan Shipyard Details

HD Hyundai Heavy Industries Ulsan shipyard key specifications:

- **Location**: Mipo Bay, Ulsan, South Korea (35.51°N, 129.42°E)
- **Area**: 1,780 acres (720 hectares)
- **Coastline**: 4 km
- **Dry docks**: 10 (lengths 260m–490m)
- **Goliath cranes**: 9 (109m tall, 900-ton capacity)
- **Annual capacity**: 50 large vessels
- **Specialty**: LNG carriers, VLCCs, container ships

B Korean Shipbuilding Production Stages

Table 28: 11-stage Korean commercial shipbuilding process with NSRP-calibrated durations. Man-hours from OECD CGT system and FMI/GSIBBS benchmarking (2005). Equipment-hours = man-hours per block / crew size.

Stage	Name	% Total	MH/Ship	Crew	Equip-hrs/Block	Equipment
0	Steel Cutting	5%	62K	8	38.8	NC plasma cutters
1	Part Fabrication	8%	99K	10	49.6	Bending machines
2	Panel Assembly	12%	149K	12	62.0	Panel line robots
3	Block Assembly	18%	223K	16	69.8	Overhead cranes
4	Block Outfitting	15%	186K	14	66.4	Manual + equipment
5	Painting	7%	87K	8	54.3	Paint booth
6	Pre-Erection	10%	124K	10	62.0	SPMTs
7	Erection	12%	149K	20	37.2	Goliath cranes
8	Quay Outfitting	10%	124K	50	2,480	Various
9	Sea Trials	3%	37K	–	72	Vessel itself
10	Delivery	–	–	–	–	–